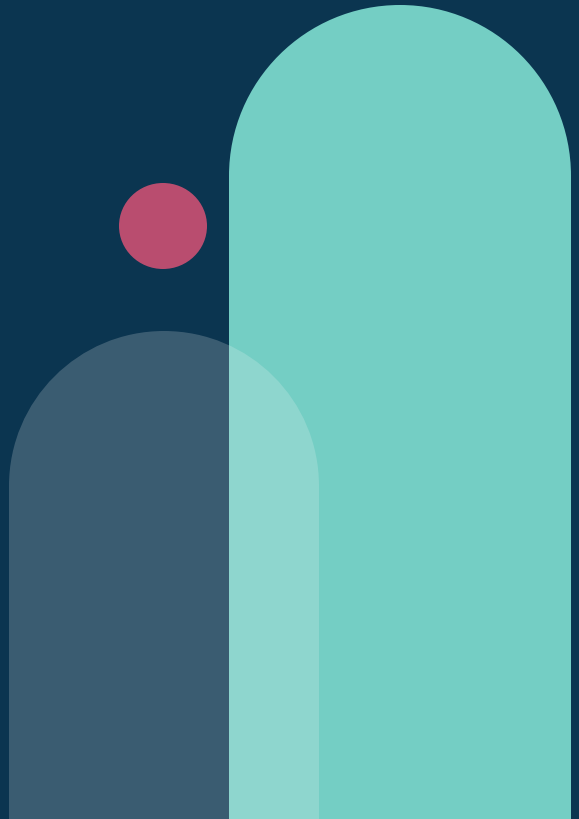
More GenAI Frameworks

01

HuggingFace Transformers



🤗 Transformers

State-of-the-art Machine Learning for **PyTorch**, **TensorFlow**, and **JAX**.

Provides APIs and tools to **download** and **train** state-of-the-art pretrained models.

Benefits:

- **Reduce compute costs** and carbon footprint.
- **Save time** and resources required for training from scratch.

Framework Interoperability:

- Support for **PyTorch**, **TensorFlow**, and **JAX**.
- Flexibility to **train in one framework** and **infer in another**.
- Export models to formats like **ONNX** and **TorchScript** for production.

Multimodal

- Image-Text-to-Text
- Visual Question Answering
- Document Question Answering
- Video-Text-to-Text

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Text-to-Image
- Image-to-Text
- Image-to-Image
- Image-to-Video
- Unconditional Image Generation
- Video Classification
- Text-to-Video
- Zero-Shot Image Classification
- Mask Generation
- Zero-Shot Object Detection
- Text-to-3D
- Image-to-3D
- Image Feature Extraction

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Feature Extraction
- Text Generation
- Text2Text Generation
- Fill-Mask
- Sentence Similarity

Audio

- Text-to-Speech
- Text-to-Audio
- Automatic Speech Recognition
- Audio-to-Audio
- Audio Classification
- Voice Activity Detection

Tabular

- Tabular Classification
- Tabular Regression
- Time Series Forecasting

Reinforcement Learning

- Reinforcement Learning
- Robotics

Other

- Graph Machine Learning

Transformers Pipeline

The pipeline() API simplifies access to pretrained models for common tasks.

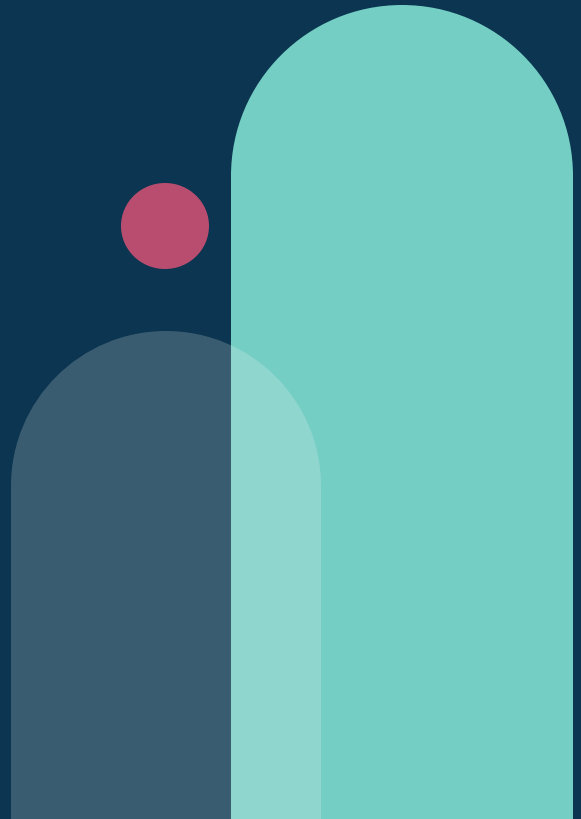
Unified API for various tasks like **text generation**, **summarization**, **translation**, etc.

Minimal code: load a model and perform inference in just **two lines**.

Task	Description	Modality	Pipeline identifier
Text classification	assign a label to a given sequence of text	NLP	pipeline(task="sentiment-analysis")
Text generation	generate text given a prompt	NLP	pipeline(task="text-generation")
Summarization	generate a summary of a sequence of text or document	NLP	pipeline(task="summarization")
Image classification	assign a label to an image	Computer vision	pipeline(task="image-classification")
Image segmentation	assign a label to each individual pixel of an image (supports semantic, panoptic, and instance segmentation)	Computer vision	pipeline(task="image-segmentation")
Object detection	predict the bounding boxes and classes of objects in an image	Computer vision	pipeline(task="object-detection")
Audio classification	assign a label to some audio data	Audio	pipeline(task="audio-classification")
Automatic speech recognition	transcribe speech into text	Audio	pipeline(task="automatic-speech-recognition")
Visual question answering	answer a question about the image, given an image and a question	Multimodal	pipeline(task="vqa")
Document question answering	answer a question about the document, given a document and a question	Multimodal	pipeline(task="document-question-answering")
Image captioning	generate a caption for a given image	Multimodal	pipeline(task="image-to-text")

02

CrewAI



What is CrewAI?

CrewAI is a cutting-edge framework for orchestrating role-playing, autonomous AI agents.

It fosters **collaborative intelligence**, enabling agents to work together to solve complex tasks.

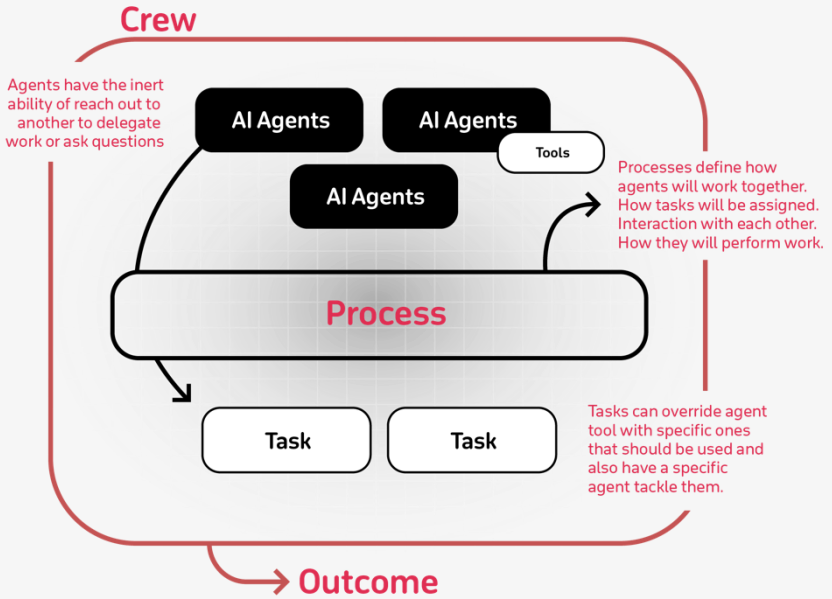
Ideal for creating systems such as:

- **Smart assistants**
- **Automated customer service**
- **Multi-agent research teams**



Key Features

Role-Based Agent Design: Customize agents with roles, goals, and tools.
Autonomous Delegation: Agents can autonomously delegate tasks and collaborate to enhance efficiency.
Flexible Task Management: Define and assign tasks dynamically.
Process-Driven Execution: Supports **sequential** and **hierarchical** task execution processes.
Works with Open Source Models: Connect to models like OpenAI or local models.
Output Management: Save outputs as files or JSON.

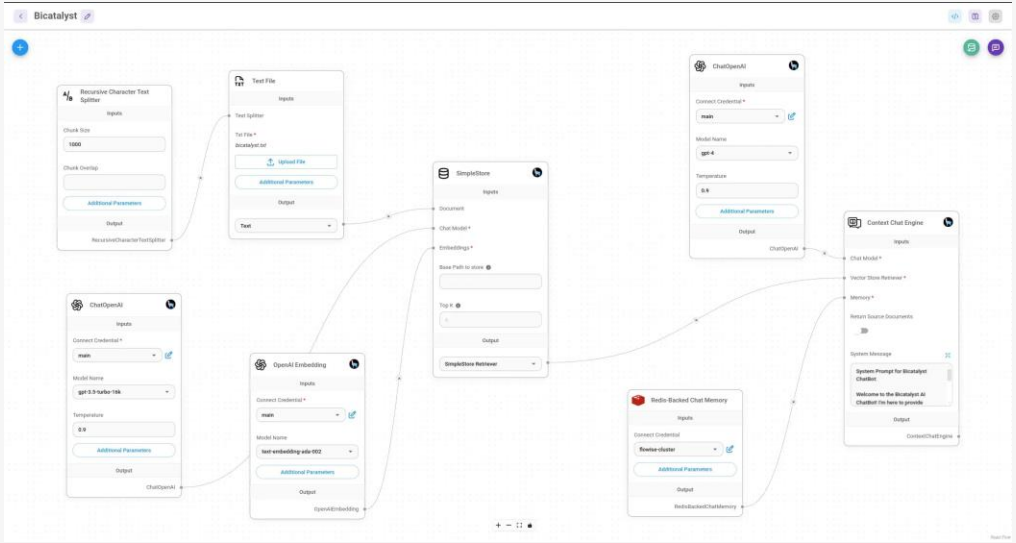


03

Flowise

Flowise: Visual Framework for LLMs

Flowise is a **no-code** visual framework for building applications powered by **Large Language Models (LLMs)**.



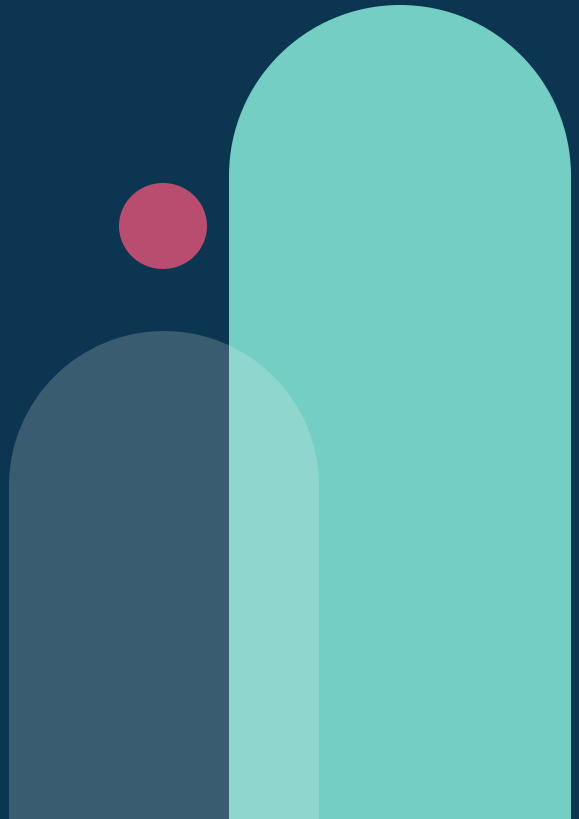
Key Features:

- **Drag-and-drop**
- **Supports multiple LLMs**
- **Integration-friendly**
- **Real-time deployment**

Perfect for **developers** and **non-technical users** looking to leverage LLMs without deep technical knowledge.

04

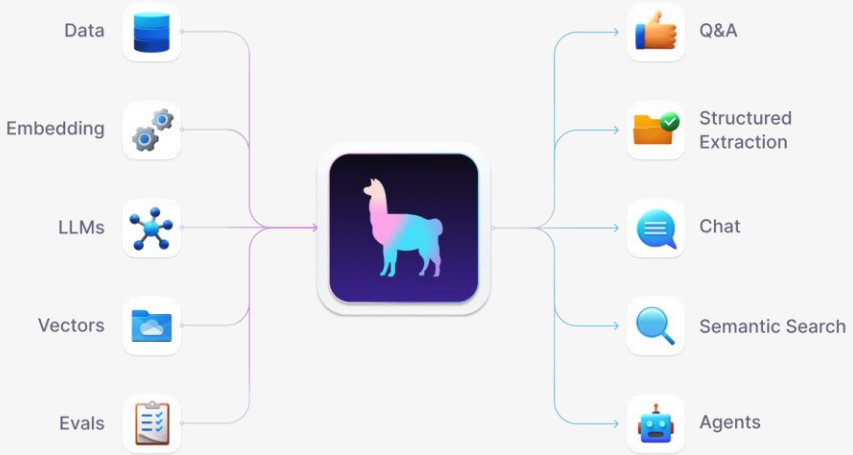
LlamaIndex



What is LlamaIndex?

LlamaIndex is a data framework designed for **LLM applications**.
Helps integrate **private data** with LLMs to enhance performance and knowledge retrieval.
Key Tools:

- **Data connectors**: Ingest data from sources like APIs, PDFs, SQL, etc.
- **Data structuring**: Create indices and graphs for efficient querying.
- **Advanced retrieval**: Retrieve context and knowledge to augment LLM outputs.
- **Seamless integrations**: Works with popular frameworks like **LangChain**, **Flask**, and **Docker**.



Why Use LlamaIndex?

High-level API: Ingest and query data in **5 lines of code**.

Customizable: Advanced users can modify connectors, indices, and retrievers.

Support for various LLMs:

OpenAI, **HuggingFace**, and other open-source models.

Ecosystem:

- **LlamaHub**: A community library for data loaders.
- **LlamaLab**: Cutting-edge AGI projects using LlamaIndex.

Persistent storage: Save and load indices to and from disk for large-scale applications.

Thanks!

Do you have any questions?

